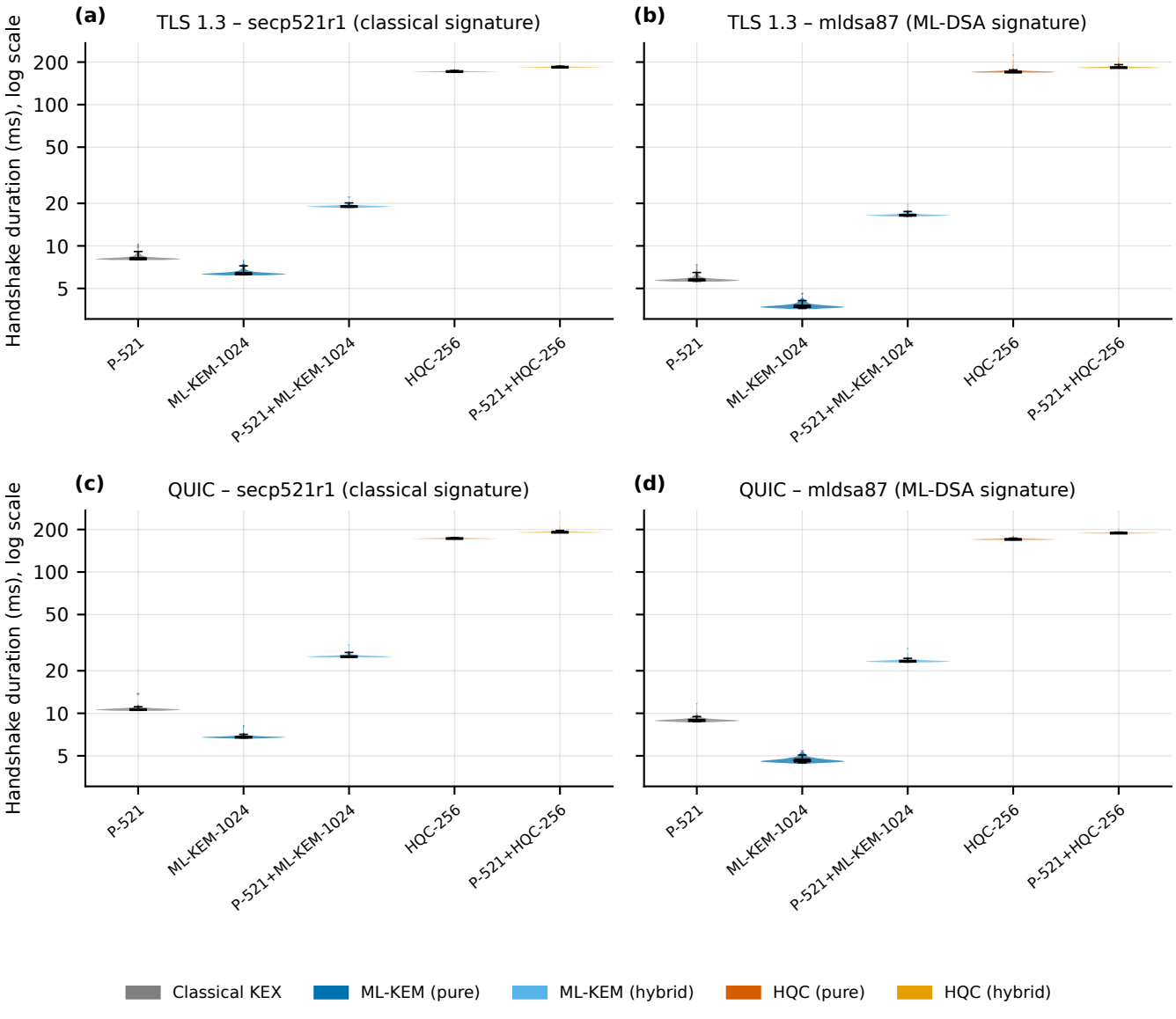

(a)
TLS 1.3 – secp521r1 (classical signature)
(b)
TLS 1.3 – mldsa87 (ML-DSA signature)
(c)
QUIC – secp521r1 (classical signature)
(d)
QUIC – mldsa87 (ML-DSA signature)
Handshake duration (ms), log scale
200
100
50
20
10
5
P-521
ML-KEM-1024
P-521+ML-KEM-1024
HQC-256
P-521+HQC-256
Classical KEX
ML-KEM (pure)
ML-KEM (hybrid)
HQC (pure)
HQC (hybrid)